

The Estimator Is the Curriculum:

Frontier Sampling and Failure Recycling for Likelihood-Based RL

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Code, logs, and proofs: <https://github.com/linjiw/curriculum-maxrl>

Abstract

Reinforcement learning with verifiable rewards spends most of its budget generating rollouts, and on hard task pools most of that budget buys nothing: under uniform prompt sampling, we measure that **65–75% of rollout groups produce zero learning signal** — every rollout fails, and success-conditioned estimators drop the group. The emerging fix is a difficulty curriculum, but current curricula are heuristics bolted on from outside: bandit scores, hand-set difficulty bands, filtering rules. We show the curriculum is already inside the estimator. For MaxRL-style success-conditioned advantages, the expected learning signal a prompt emits from a group of N rollouts has a closed form, $\mathbb{E}[\sum_i |w_i|] = 2(\text{pass}@N - \text{pass}@1)$ — a compute-indexed zone-of-proximal-development functional, maximal on prompts solvable within N attempts but not within one, peaking at pass rate $\approx \ln N/N$. Sampling prompts by this quantity (a decayed Thompson posterior makes it practical) turns the rollout budget N you already chose into the curriculum knob, with no new difficulty hyperparameters; published learnability curricula fall out as its $N=2$ slice. The same algebra draws a hard boundary: no sampling rule can rescue an all-fail prompt, because allocation only redistributes signal that exists. We therefore add a second, complementary mechanism — **failure recycling**: a failed trajectory is a verified success for the goal it actually reached, and relabeling dead groups (with the prompt’s goal conditioning rewritten) provably yields the maximum-likelihood gradient of the relabeled task, exact when conditional laws match. On testbeds spanning exact-gradient chains, a 1.26M-parameter maze transformer, Gymnasium control, and a 360M LLM on GSM8K, the combined schedule beats a true-pass-rate **oracle** allocator (0.890 vs. 0.851 AUC), turns a frontier-heavy regime from unlearnable to solved ($0 \rightarrow 0.98$ at equal compute), grows $\text{pass}@k$ coverage where GRPO’s collapses, and needs up to $11\times$ fewer inference samples to reach target coverage. A pre-registered LLM-scale 2×2 confirms the safety half of the claim: the identical curriculum that trains stably under MaxRL *degrades* GRPO — curricula amplify objective-level pathologies, so the estimator underneath decides whether a curriculum is safe at all.

1 Introduction

Post-training language models with verifiable rewards has a cost structure unlike ordinary supervised learning: the data is free, but *rollouts* are not. Every optimization step is preceded by sampling N completions per prompt, and generation dominates wall-clock (86% of step time in our LLM runs). What that compute buys depends entirely on where it lands. On a prompt the model has mastered, N successes carry no contrast and no gradient. On a prompt beyond the model’s reach, N failures carry no successes — and for the success-conditioned estimators of likelihood-based RL, the group is dropped entirely. We measure the combined waste at 65–75% of groups under uniform sampling. The field’s response is difficulty curricula — but the current generation derives its sampling rule from outside the learner: UCB bandits over difficulty buckets [2], target-difficulty controllers [3], category bandits [4], rejection rules [5], or dead-prompt filtering [6, 7]. Each adds machinery and hyperparameters to estimate, in effect, *where learning is possible right now*.

Our starting observation is that likelihood-based RL already computes this. MaxRL [1] showed that standard RL on binary rewards optimizes a first-order approximation of maximum likelihood, and that normalizing advantages by the group’s success count K instead of its size N recovers a truncated-ML objective. Reading their weight-function view raises a question with a non-obvious answer: *if the objective already pours gradient into hard prompts, is there anything left for a curriculum to do?* The answer is sharply yes, for a structural reason: **weights act on prompts after they are sampled, and only when at least one**

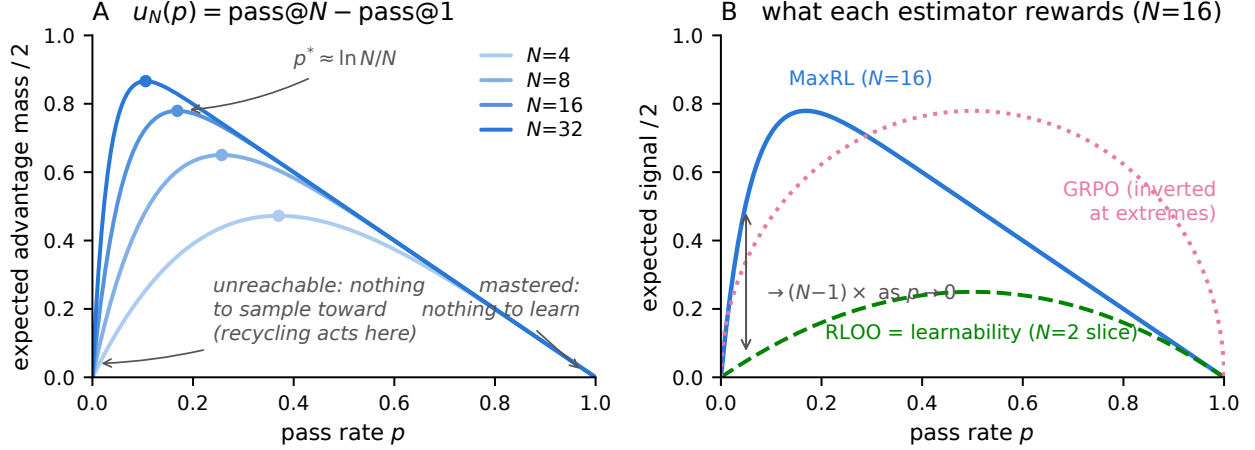


Figure 1: **The estimator is the curriculum.** *Left:* the derived utility $u_N(p) = (1 - (1 - p)^N) - p$ — the estimator’s expected advantage mass (Prop. 1) — for growing rollout budgets N . The peak $p^* \approx \ln N/N$ walks toward harder prompts as compute grows; both tails are dead zones the weight function cannot reach. *Right:* at $N=16$, MaxRL concentrates $\rightarrow (N-1) \times$ more expected signal than RLOO on frontier prompts (Prop. 2); RLOO’s profile is exactly the published “learnability” objective — the $N=2$ slice.

rollout succeeds. However the weight function is shaped, it cannot rescue a group that came back all-fail — dropping that group is precisely what makes the estimator unbiased — and it cannot un-spend the compute burned re-confirming mastered prompts. The estimator’s blind spots define the curriculum’s job description.

Taking that sentence literally produces this paper. Our contributions:

1. **The curriculum is a theorem, not a heuristic** (§3): the expected advantage mass of a prompt is exactly $2(\text{pass}@N - \text{pass}@1)$, a derived ZPD functional whose band location, width, and compute-scaling are set by N alone. Learnability curricula are its $N=2$ slice; optimal rollout allocation is water-filling on $p(1 - p)^N$.
2. **A practical teacher** (§4): Thompson sampling on a decayed Beta posterior with the derived utility; the estimator is untouched, so its unbiasedness carries over.
3. **Failure recycling with an exactness guarantee** (§5): relabeling dead groups to achieved sub-goals yields the ML gradient of the relabeled task; exact when conditional laws match, measured indistinguishable from fresh groups (cosine 0.956 vs. 0.958), with two contracts whose violation we price.
4. **A three-channel account with a safety rule** (§6–7): allocation is bounded by an oracle ceiling; creation breaks it; the objective underneath decides whether either is safe — confirmed at LLM scale on a pre-registered prediction.
5. **An honest evidence discipline:** pre-registered predictions, documented negatives with diagnoses, one retraction, and a two-round adversarial audit tracing every number to a log or proof.

2 Background: MaxRL in three lines

For binary rewards, maximum likelihood maximizes $\mathbb{E}[\log p_\theta(\text{success} \mid x)]$. MaxRL expands $\log p$ as a Maclaurin series in $(1 - p)$ and truncates at order T , giving a compute-indexed family $J^{(T)}$ interpolating REINFORCE ($T=1$) to exact ML ($T \rightarrow \infty$); the practical estimator sets per-rollout advantages $w_i = r_i/K - 1/N$ with all-fail groups dropped, which is unbiased for the $T = N-1$ objective. Its weight function grows as $p \rightarrow 0$: the objective is an *implicit, gradient-level* curriculum. Our work is the data-level complement: the same algebra, read as a sampling rule plus a recycling rule.

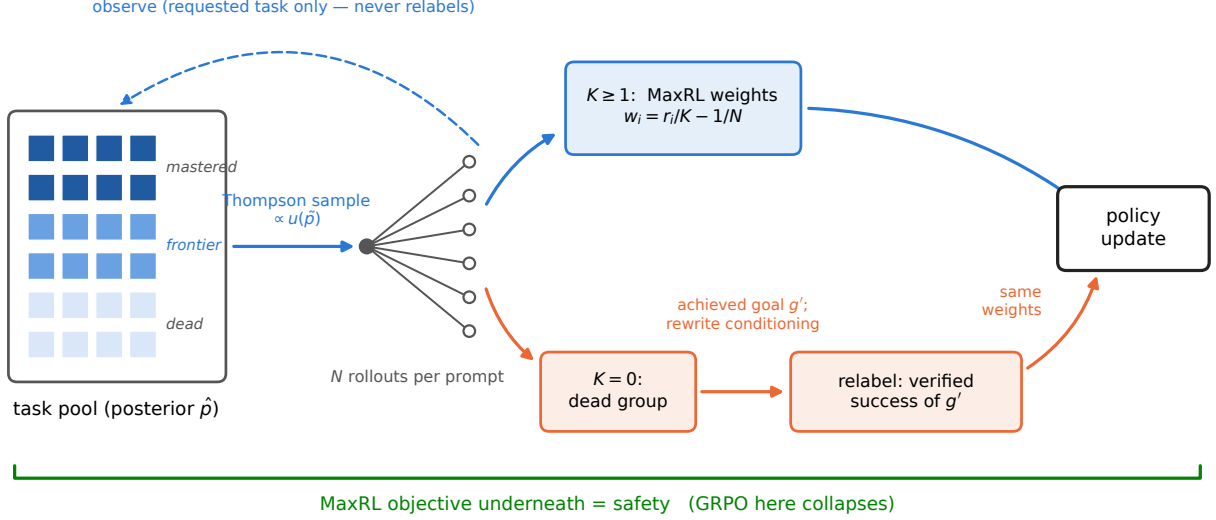


Figure 2: **One FrontierMax step.** A Thompson-sampled posterior allocates rollouts by the derived utility; live groups train with unmodified MaxRL weights; dead groups are relabeled to the sub-goal they actually achieved (conditioning rewritten) and train as verified successes. The posterior observes requested-task outcomes only.

3 The estimator is the curriculum

Proposition 1 (Advantage mass; MC-verified). *For a prompt with pass rate p and N i.i.d. rollouts under the practical MaxRL weights,*

$$\mathbb{E} \left[\sum_i |w_i| \right] = 2(\text{pass@}N(p) - \text{pass@}1(p)) = 2((1 - (1 - p)^N) - p).$$

Interpretation. *The estimator’s expected learning signal on a prompt is twice the probability that the prompt is solvable within N attempts but not within one. That sentence is a curriculum: zero on mastered prompts, zero on unreachable ones, maximal at $p^* = 1 - N^{-1/(N-1)} \approx \ln N/N$ — a zone of proximal development whose center, width, and compute-scaling are set by the one parameter you already chose.*

Proposition 2 (Frontier amplification). *As $p \rightarrow 0$, the ratio of MaxRL’s expected signal to RLOO’s ($2p(1 - p)$ exactly) tends to $N - 1$.*

Interpretation. *Concentrating sampling on the frontier only helps if the estimator can extract signal there; GRPO’s variance-normalized weights invert the profile, which §7 shows becoming an active failure under a curriculum.*

Proposition 3 (Allocation). *The rollout allocation maximizing total expected signal is greedy water-filling on the marginal $p(1 - p)^N$ — the probability that the next rollout is a group’s first success.*

One identity, three literatures: learnability curricula [5, 9] are the $N=2$ slice of u_N ; DAPO-style dynamic sampling and GRESO are its avoidance shadow (cull where $u \approx 0$); HER-style relabeling is its creation complement (manufacture $K > 0$ where $u = 0$ identically; §5).

4 FrontierMax: the schedule

A decayed Beta posterior (decay 0.7) tracks each prompt’s pass rate from observed group outcomes only — never from relabeled successes (violating this inflates the posterior: $\hat{p}$ 0.81 vs. eval 0.47). Thompson sampling draws $\tilde{p}$ and samples prompts $\propto u(\tilde{p})^\gamma$ with a 10% uniform floor; $\gamma \approx 4$ when tasks share skills

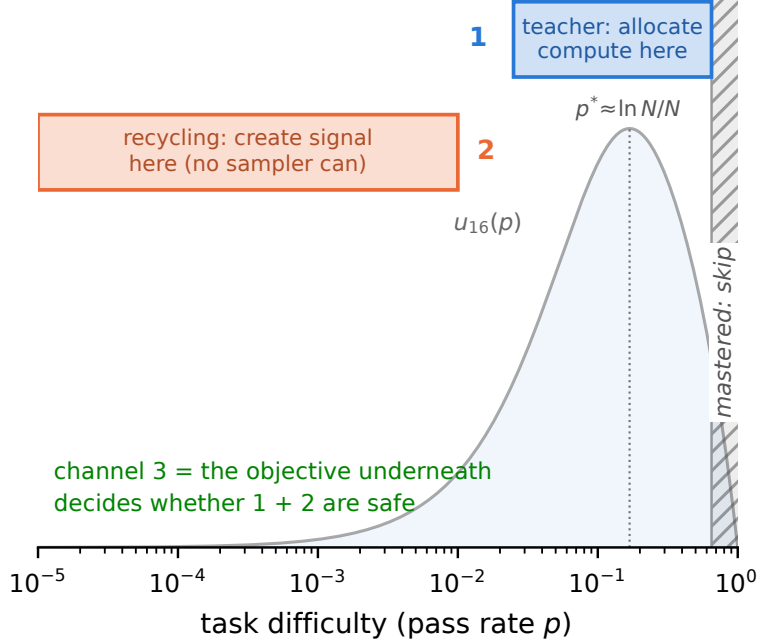


Figure 3: **Three channels.** The teacher allocates compute to the frontier band; recycling creates signal in the dead zone no sampler can reach; the objective underneath decides whether either is safe.

(learning compounds — validated on chains, and its non-transfer to broad pools was predicted in advance by a compounding ODE model), $\gamma = 1$ otherwise. Live groups train with unmodified MaxRL advantages. Honest knob inventory: decay, floor, and γ exist, with validated defaults; what is *derived* rather than tuned is the difficulty band itself — location, width, and N -scaling — which is exactly where competing curricula spend their hyperparameters.

5 Failure recycling

Proposition 4 (Hindsight exactness). *Relabeling a dead group to the sub-goal g' its trajectories actually achieved, rewriting the goal conditioning, and applying the same success-conditioned weights yields the ML gradient of task g' under the conditional law shifted by the original sampling; the two coincide exactly when the conditional laws match.*

Interpretation. *Measured: per-group cosine to the true relabeled-task gradient 0.956, vs. 0.958 for fresh unbiased groups; the mean relabeled gradient reaches cosine 1.000.*

Two contracts keep practice on the proof’s side: **(1) exactness** — a relabeled success must be a true success of the relabeled task under the environment’s own verifier (never an LLM judge); **(2) conditioning** — goal-embedding trajectories must have their conditioning rewritten to the achieved goal. Skipping (2) turns the exact gradient into noise: on our gridworld, hindsight without the rewrite scores below teacher-only ($0.600 < 0.658$); with it, it leads (0.703). Recycling is the only mechanism here that *creates* signal, and its value is regime-dependent in a way we can state precisely: the gain is proportional to how much a relabeled skill can compound — +0.22 AUC on fixed task pools, +0.01 on one-shot task streams. Task-set fixedness is the single most important regime variable we found.

6 Three channels, one safety rule

The **teacher** avoids waste — bounded by an oracle ceiling (a perfect allocator collects only 0.4% more advantage mass than our posterior). **Recycling** creates signal — the only channel that breaks that ceiling.

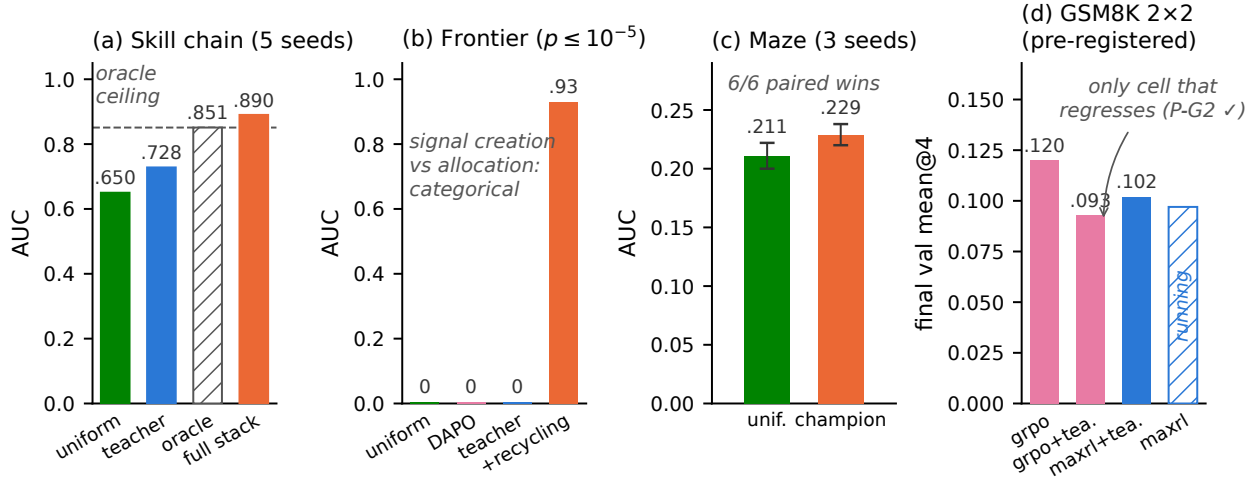


Figure 4: **The ladder.** (a) Exact-gradient chains: the full stack beats the true-pass-rate oracle. (b) Frontier-heavy regime: allocation scores exactly zero; creation solves it. (c) Maze at matched wall-clock: 6/6 paired seed wins. (d) GSM8K 2x2: the identical teacher helps MaxRL and hurts GRPO (pre-registered P-G2).

The **objective** underneath decides whether either is safe. One line: *the teacher allocates, hindsight creates, the objective decides whether either is safe.*

7 Experiments: an escalating ladder

7.1 Exact-gradient skill chains (CPU, 5 seeds). Uniform 0.650 $\rightarrow$ teacher 0.728 $\rightarrow$ full stack **0.890 $\pm$ 0.002** — above the true-pass-rate oracle allocator (0.851) and above a sharper $\gamma=4$ oracle (0.884).

Takeaway. Allocation saturates; creation breaks the ceiling.

7.2 Frontier-heavy regime (max pool pass rate 10^{-5}). Uniform, DAPO, and the plain teacher score **exactly 0.00**; teacher+recycling reaches **0.98 final (0.93 AUC)** — relabeling invents the curriculum below the pool, ignites within ~ 400 groups, then goes silent.

Takeaway. Where there is nothing to sample toward, only signal creation works — categorically, not marginally.

7.3 Maze transformer at matched wall-clock (1.26M params, 13 levels, ~ 30 runs, 3 seeds). Champion (teacher + dense recycling) 0.252 ± 0.005 final / 0.229 ± 0.009 AUC vs. uniform 0.230 ± 0.015 / 0.211 ± 0.011 ; paired deltas positive 6/6 seeds; 22–35% more optimization steps per GPU-hour. Safety: the identical teacher grows pass@8 under MaxRL in every seed ($0.316 \rightarrow 0.348$) while GRPO’s coverage decays in every seed ($0.308 \rightarrow 0.271$); the teacher-arm run amplifies the collapse ($0.332 \rightarrow 0.269$; single-seed arm). Inference currency: $1.2 \times / 2.7 \times / 11 \times$ fewer samples than GRPO to reach target coverage at levels 2/3/5 (honest $0.5 \times$ reversal at level 4).

Takeaway. The gains survive real gradients; the safety warning is real; coverage is the currency that sees both.

7.4 Gymnasium control (MountainCar). Flag-only training: 0.000 — the classic exploration wall. The corrected 10-seed transition-matched study (paired bootstrap): flag pass uniform $0.058 \pm 0.079 \rightarrow$ teacher $0.664 \pm 0.232 \rightarrow$ full stack **0.848 $\pm$ 0.058**; per-bin policy parameters stay at 0.000.

Takeaway. Curricula operate through shared parameters, or not at all.

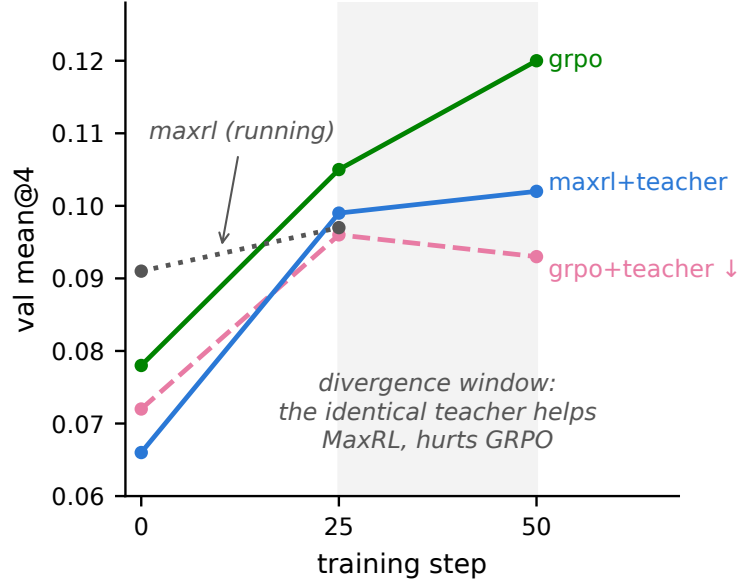


Figure 5: **GSM8K 2×2 (pre-registered)**. The identical teacher helps MaxRL and hurts GRPO; GRPO+teacher is the only cell that regresses in its second half — the maze safety reversal at LLM scale.

7.5 LLM scale: GSM8K 2×2, pre-registered (SmolLM2-360M, $N=16$, one A10G). Predictions committed before any cell finished. **P-G2, the riskiest prediction, confirmed:** GRPO+teacher is the only cell that regresses in its second half (val accuracy .096 $\rightarrow$.093, pass@4 .193 $\rightarrow$.181) while plain GRPO climbs monotonically (.078 $\rightarrow$.105 $\rightarrow$.120) — with the teacher verifiably steering (sampled-dead fraction driven to 0.48 vs. the ~ 0.65 population rate). MaxRL+teacher trains stably to its best value (.066 $\rightarrow$.102); the uniform-MaxRL cell finishes at .108, making P-G1 (a teacher gain for MaxRL) an honest *null at this budget* — predicted in advance by the posterior-starvation analysis below. A pass@ k sweep on the final checkpoints sharpens the safety read: the teacher’s deficit under GRPO *widens monotonically with k* (−.008 at $k=1$ to −.024 at $k=16$) — curriculum damage lives in coverage, as the maze predicted, though each individual delta is within single-seed noise. Beyond the predictions: the teacher’s posterior *learns* real difficulty from ~ 1 visit per prompt ($\rho = -0.17$ against independent 7B-model solve-rate annotations, $p \approx 10^{-17}$) but its allocation is *posterior-starved* — 3,200 draws over 7,473 prompts leave Thompson sampling near-uniform. And the GRPO degradation is not token-entropy collapse (GRPO+teacher retains *more* entropy): the damage is distributional, visible only in coverage. Single seed; small model; the pre-registered outcomes were orderings and signs, and both landed.

Takeaway. The safety half of the thesis transfers to LLM RLVR; the allocation half needs tiers or longer runs — which is how the next experiment is designed.

7.6 Countdown 2×2 (pre-registered): two honest nulls and a new phenomenon. The first exact-verifier hindsight experiment in RLVR ran in production (12 relabeled groups/step, 72–89% yield). Both positive predictions returned null, with an exact diagnosis: the uniform baseline reached 94/76/94% of the random-guesser bound implied by the pool’s simplified construction — no headroom for allocation or creation to matter. What the run found instead is new: **recycling-induced sharpening** — hindsight arms improved mean@16 while *losing* pass@16 coverage (tier-1: .571 vs. .645 for the no-relabel baseline). Recycled off-target successes concentrate the policy on reachable outputs at the expense of exploration breadth: the creation channel has its own safety trade-off, mirroring §7.3’s weight-induced collapse — data-induced rather than weight-induced, and to our knowledge unreported (no prior work measures pass@ k under hindsight in RLVR). Our algebra predicts it: a relabel to a self-achieved value is a task at $p \approx 1$, where $u(p)$ says training signal is zero — the softmax pays for those updates out of the exploration tail. *Takeaway: recycling needs an*

admission rule, and the estimator already provides it — the same derived utility that schedules sampling should gate the relabel destination.

8 Related work

Difficulty-adaptive RLVR sampling is bucket- or scalar-level and heuristic-scored: DUMP (UCB over buckets), AdaRFT (scalar controller over precomputed labels), SEC (category bandit), threshold curricula. LILO is the closest per-prompt method — rejection sampling by $\hat{p}(1 - \hat{p})$, which Prop. 1’s corollary identifies as the $N=2$ slice of our derived utility. DAPO’s dynamic sampling and GRESO cull dead prompts — avoidance without allocation or creation. PKPO and Pass@ k -training modify the objective toward pass@ k but keep uniform sampling. Hindsight for LLMs (AgentHER, HSL) relabels agent trajectories with an LLM judge. The nearest neighbor is concurrent: LfH [8] brings hindsight relabeling into GRPO for VLA post-training — a VLM proposes one shared hindsight instruction per failed group and rescores the group under it; $5\times$ sample efficiency on LIBERO-PRO. LfH’s relabeler is a VLM judge (their stated limitation: relabel noise), and its evaluation never measures pass@ k — precisely the currency where we find recycling’s hidden cost. Exact-verifier relabeling with a correctness guarantee, and the coverage accounting of recycling, remain, to our knowledge, ours. No prior work (i) derives the sampling utility from the estimator’s own expected signal, (ii) couples it with success-conditioned weights it provably matches, and (iii) adds a signal-creation channel with an exactness guarantee.

9 Limitations and honest negatives

Deep frontiers at fixed budget remain uncrossed on the maze, and a $4\times$ duration run *refuted* our own duration hypothesis (the stall is a per-step-competence ceiling; depth needs capacity, not more schedule). γ -concentration did not transfer from chains to the maze — predicted in advance by the compounding ODE model. Adaptive truncation order underperformed fixed T . Feeding relabeled successes to the teacher’s posterior inflates it — dropped. An early GSM8K claim of teacher steering was retracted when review found the sampler epoch-frozen; the fixed run confirmed the effect properly. Recycled-gradient exactness is a property of the environment’s relabel map; noisier verifiers will land between our $+0.22$ (fixed pools) and $+0.01$ (one-shot) endpoints. LLM results are single-seed at 360M; the $2\times 2\times 2$ and multi-seed replications are the active work.

Reproducibility

Every proposition has a Monte-Carlo verification script; every experiment writes JSON/JSONL artifacts committed to the repository; the documentation passed a two-round adversarial audit. The framework is a dependency-light package (`frontier_rl`, numpy-only core) with adapters for LLM pools (verl), gym control, IsaacLab parallel sim, and flow-policy VLAs.

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